

Constant factor to $(1+\epsilon)$ -approximation for structured matrix learning

Abstract

We study hypothesis selection in the matrix–vector query model. A family $\mathcal{F} = \{\mathbf{A}_1, \dots, \mathbf{A}_N\} \subset \mathbb{R}^{n \times n}$ is known, an unknown $\mathbf{A} \in \mathbb{R}^{n \times n}$ is accessible only through the products $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ and $\mathbf{x} \mapsto \mathbf{A}^\top \mathbf{x}$, and the goal is to identify a member of $\mathcal{F}$ that is nearly as close to $\mathbf{A}$ in Frobenius norm as the best one. Estimating the N distances separately costs $O(\epsilon^{-2} \log N)$ queries. We show that, given a warm start — any constant-factor approximation to $\mathbf{A}$, not necessarily drawn from $\mathcal{F}$ — only

$$O(\epsilon^{-2} \sqrt{\log N})$$

queries are needed, a quadratic improvement in the dependence on $\log N$, with no dependence on the dimension n .

The saving comes from using both sides of the matrix. Writing k for the number of queries spent, a randomized rangefinder first deflates the residual between $\mathbf{A}$ and the warm start, leaving a remainder that is *spectrally flat*: its spectral norm is a $1/\sqrt{k}$ fraction of its Frobenius norm. A single shared Gaussian quadratic sketch then estimates all N inner products against that remainder at once. Flatness is what makes the sketch concentrate as though it drew k^2 samples rather than k , so a union bound over N hypotheses is paid at $k = \Theta(\sqrt{\log N})$ instead of $\Theta(\log N)$. Deflation is the step that requires transpose access, and we show this is unavoidable: in any bounded-precision model, algorithms restricted to one-sided queries need $\tilde{\Omega}(\log N)$ queries even when handed a warm start.

1 Model and main result

Throughout, $\mathcal{F} = \{\mathbf{A}_1, \dots, \mathbf{A}_N\} \subset \mathbb{R}^{n \times n}$ is known and $\mathbf{A} \in \mathbb{R}^{n \times n}$ is unknown. The algorithm may issue *right queries* $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ and *left queries* $\mathbf{x} \mapsto \mathbf{A}^\top \mathbf{x}$ with arbitrary real query vectors $\mathbf{x} \in \mathbb{R}^n$, and observes the full output vector exactly. Each query vector may depend on the answers to all previous queries. The query complexity is the total number of matrix–vector products issued; all other computation (in particular, on the known family $\mathcal{F}$) is free. We write

$$d_i := \|\mathbf{A} - \mathbf{A}_i\|_F, \quad \text{OPT} := \min_{i \in [N]} d_i, \quad i^{\text{opt}} \in \arg \min_{i \in [N]} d_i. \quad (1)$$

The goal is to output an index $i^* \in [N]$ with $d_{i^*} \leq (1 + \epsilon) \text{OPT}$.

We use $\|\cdot\|_2$ for the spectral norm, $\|\cdot\|_*$ for the nuclear norm, $\sigma_1(\mathbf{X}) \geq \sigma_2(\mathbf{X}) \geq \dots$ for singular values, and $[\mathbf{X}]_r$ for a best rank- r approximation of $\mathbf{X}$ in Frobenius norm. The inner product is $\langle \mathbf{X}, \mathbf{Y} \rangle = \text{tr}(\mathbf{X}^\top \mathbf{Y})$.

We assume throughout that a *warm start* is available.

Assumption 1.1 (Warm start). *A matrix $\bar{\mathbf{A}}_0 \in \mathbb{R}^{n \times n}$ with known entries, not necessarily in $\mathcal{F}$, is given, together with a constant $\gamma \geq 1$ such that $\|\mathbf{A} - \bar{\mathbf{A}}_0\|_F \leq \gamma \text{OPT}$.*

The constant γ need not be tight; only the bound is used. Note that ?? is nontrivial exactly when $\text{OPT} > 0$: if $\text{OPT} = 0$ then $\bar{\mathbf{A}}_0 = \mathbf{A}$ and the problem is solved offline.

Theorem 1.2 (Main). *Let $\varepsilon \in (0, 1]$ and $\delta \in (0, 1/2)$, and let ?? hold. There is an algorithm making at most*

$$O\left(\gamma^4 \varepsilon^{-2} \sqrt{\log(N/\delta)} + \log(1/\delta)\right)$$

queries that outputs i^ with $d_{i^*} \leq (1 + \varepsilon)\text{OPT}$ with probability at least $1 - \delta$. The query count is independent of n .*

For constant γ and δ this reads $O(\varepsilon^{-2} \sqrt{\log N})$, the form quoted in the abstract. The algorithm uses only two rounds of queries: all right queries are made first, with vectors drawn independently of $\mathbf{A}$, and the left queries follow, with vectors spanning the range of the first round's output.

The argument has three moving parts. ?? reduces selection to estimating N inner products against the residual, and shows how accurate those estimates must be. ?? gives the estimator and its analysis; this is where the $\sqrt{\log N}$ appears. ?? combines the two. ?? explains why two-sided access is necessary (??) and why deflation is (??).

2 Preliminaries

We need two standard tools: Hanson–Wright concentration for Gaussian quadratic forms, and the spectral-norm guarantee of the randomized rangefinder.

Lemma 2.1 (Hanson–Wright, averaged form [Hanson1971; rudelson2013hanson]). *Let $\mathbf{N} \in \mathbb{R}^{n \times n}$ be fixed and let $\mathbf{y}_1, \dots, \mathbf{y}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ be independent. Then for every $t \geq 0$,*

$$\Pr \left[\left| \frac{1}{k} \sum_{b=1}^k \mathbf{y}_b^\top \mathbf{N} \mathbf{y}_b - \text{tr } \mathbf{N} \right| \geq t \right] \leq 2 \exp \left(-\Omega \left(\min \left\{ \frac{kt^2}{\|\mathbf{N}\|_F^2}, \frac{kt}{\|\mathbf{N}\|_2} \right\} \right) \right). \quad (2)$$

We emphasize that ?? does not require $\mathbf{N}$ to be symmetric or positive semidefinite. Indeed, $\mathbf{y}^\top \mathbf{N} \mathbf{y} = \mathbf{y}^\top \mathbf{N}_{\text{sym}} \mathbf{y}$ and $\text{tr } \mathbf{N} = \text{tr } \mathbf{N}_{\text{sym}}$ for the symmetric part $\mathbf{N}_{\text{sym}} := \frac{1}{2}(\mathbf{N} + \mathbf{N}^\top)$, so the symmetric case applies verbatim; and since $\|\mathbf{N}_{\text{sym}}\|_F \leq \|\mathbf{N}\|_F$ and $\|\mathbf{N}_{\text{sym}}\|_2 \leq \|\mathbf{N}\|_2$ by the triangle inequality, replacing $\mathbf{N}_{\text{sym}}$ by $\mathbf{N}$ in the exponent of ?? only weakens the bound. This is used with $\mathbf{N}_i = (\mathbf{B}_i^H)^\top \mathbf{R}$ in ??, which is not symmetric.

Lemma 2.2 (Deflation via the randomized rangefinder [HMT11]). *Let $\mathbf{M} \in \mathbb{R}^{n \times n}$, let $\boldsymbol{\Omega} \in \mathbb{R}^{n \times k}$ have i.i.d. $\mathcal{N}(0, 1)$ entries with $k \geq 4$, and let $\mathbf{Q} = \text{orth}(\mathbf{M}\boldsymbol{\Omega})$. Then with probability $1 - e^{-\Omega(k)}$,*

$$\|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{M}\|_2 = O\left(\frac{\|\mathbf{M}\|_F}{\sqrt{k}}\right). \quad (3)$$

In words: sketching with k Gaussian vectors leaves a residual whose spectral norm is a $1/\sqrt{k}$ fraction of the Frobenius norm of $\mathbf{M}$ — the residual is *spectrally flat*. This is the only property of the rangefinder we use.

Proof. The claim is trivial for $k = O(1)$, since $\|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{M}\|_2 \leq \|\mathbf{M}\|_F$ always; so assume k exceeds any fixed constant. Write $\sigma_j := \sigma_j(\mathbf{M})$ and split k into a target rank $\rho := \lfloor k/2 \rfloor - 1$ and an oversampling parameter $p := k - \rho$, so that $\rho \leq p$ and $p = \Theta(k)$. The deviation bound for the Gaussian rangefinder [HMT11], applied with $t = e$ and $u = \sqrt{k}$, gives with probability $1 - e^{-\Omega(k)}$

$$\|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{M}\|_2 \leq \left(1 + e\sqrt{12\rho/p} + \frac{e^2\sqrt{k}\sqrt{\rho+p}}{p+1}\right)\sigma_{\rho+1} + \frac{e^2\sqrt{\rho+p}}{p+1}\left(\sum_{j>\rho}\sigma_j^2\right)^{1/2}.$$

Since $\rho \leq p$ and $\rho + p = k = O(p)$, the coefficient of $\sigma_{\rho+1}$ is $O(1)$ and that of the tail is $O(1/\sqrt{k})$. The tail is at most $\|\mathbf{M}\|_F$, while $\rho + 1 = \Theta(k)$ forces $\sigma_{\rho+1} \leq \|\mathbf{M}\|_F/\sqrt{\rho+1} = O(\|\mathbf{M}\|_F/\sqrt{k})$. Both terms are therefore $O(\|\mathbf{M}\|_F/\sqrt{k})$. $\square$

We record two elementary facts used repeatedly. For any $\mathbf{X}$ with $\text{rank}(\mathbf{X}) \leq r$ we have $\|\mathbf{X}\|_* \leq \sqrt{r}\|\mathbf{X}\|_F$. For any $\mathbf{X}$ and any $r \geq 0$, the tail $\mathbf{X} - [\mathbf{X}]_r$ satisfies $\|\mathbf{X} - [\mathbf{X}]_r\|_2 = \sigma_{r+1}(\mathbf{X}) \leq \|\mathbf{X}\|_F/\sqrt{r+1}$.

3 Reduction: selection from approximate linear functionals

Fix a *pivot* $\bar{\mathbf{A}} \in \mathbb{R}^{n \times n}$ whose entries are known to the algorithm (it need not lie in $\mathcal{F}$), and write

$$\mathbf{B} := \mathbf{A} - \bar{\mathbf{A}}, \quad \mathbf{B}_i := \mathbf{A}_i - \bar{\mathbf{A}} \quad (i \in [N]).$$

Expanding $d_i^2 = \|\mathbf{B} - \mathbf{B}_i\|_F^2$ gives

$$d_i^2 = \|\mathbf{B}\|_F^2 - 2\langle \mathbf{B}, \mathbf{B}_i \rangle + \|\mathbf{B}_i\|_F^2. \quad (4)$$

The term $\|\mathbf{B}\|_F^2$ is unknown but common to all i , and $\|\mathbf{B}_i\|_F^2$ is computable offline. So selection reduces to estimating $\langle \mathbf{B}, \mathbf{B}_i \rangle$ for all $i \in [N]$. The following lemma guarantees that, for a sufficiently accurate pivot and with sufficiently accurate estimates, we can easily obtain a near-optimal approximation.

Lemma 3.1 (Selection from relatively accurate functionals). *Let $\varepsilon \in (0, 1]$ and $\gamma \geq 1$, and let $\bar{\mathbf{A}}$ be a pivot with $\|\mathbf{B}\|_F = \|\mathbf{A} - \bar{\mathbf{A}}\|_F \leq \gamma \text{OPT}$. Set*

$$\varepsilon_0 := \frac{\varepsilon}{2\gamma(1+\gamma)} \in (0, 1]. \quad (5)$$

Suppose $\hat{\varphi}_1, \dots, \hat{\varphi}_N$ satisfy

$$|\hat{\varphi}_i - \langle \mathbf{B}, \mathbf{B}_i \rangle| \leq \varepsilon_0 \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F \quad \text{for every } i \in [N], \quad (6)$$

and let $i^ \in \arg \min_i (\|\mathbf{B}_i\|_F^2 - 2\hat{\varphi}_i)$. Then $d_{i^*} \leq (1 + \varepsilon) \text{OPT}$.*

Proof. Write $\eta_i := \varepsilon_0 \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F$ for the error bound in ??, and $s_i := \|\mathbf{B}_i\|_F^2 - 2\hat{\varphi}_i$ for the score minimized by i^* .

Step 1: selection. By ??, every $i \in [N]$ satisfies the identity $d_i^2 = \|\mathbf{B}\|_F^2 + s_i + 2(\hat{\varphi}_i - \langle \mathbf{B}, \mathbf{B}_i \rangle)$. Applying it at $i = i^*$ and using ??, then the minimality $s_{i^*} \leq s_{i^{\text{opt}}}$, then the identity again at $i = i^{\text{opt}}$ and ?? once more,

$$d_{i^*}^2 \leq \|\mathbf{B}\|_F^2 + s_{i^*} + 2\eta_{i^*} \leq \|\mathbf{B}\|_F^2 + s_{i^{\text{opt}}} + 2\eta_{i^*} \leq d_{i^{\text{opt}}}^2 + 2\eta_{i^{\text{opt}}} + 2\eta_{i^*} = \text{OPT}^2 + 2\eta_{i^{\text{opt}}} + 2\eta_{i^*}. \quad (7)$$

The unknown quantity $\|\mathbf{B}\|_F^2$ cancels; only differences of the scores s_i are used.

If $\|\mathbf{B}\|_F = 0$ then $\eta_i = 0$ for every i and ?? gives $d_{i^*} \leq \text{OPT}$. Assume henceforth $\|\mathbf{B}\|_F > 0$, so also $\text{OPT} > 0$.

Step 2: the quadratic. The triangle inequality gives $\|\mathbf{B}_i\|_F \leq d_i + \|\mathbf{B}\|_F$, so ?? becomes

$$d_{i^*}^2 \leq \text{OPT}^2 + 2\varepsilon_0\|\mathbf{B}\|_F(\|\mathbf{B}_{i^*}\|_F + \|\mathbf{B}_{i^{\text{opt}}}\|_F) \leq \text{OPT}^2 + 2\varepsilon_0\|\mathbf{B}\|_F(d_{i^*} + \text{OPT} + 2\|\mathbf{B}\|_F).$$

Write $a := 2\varepsilon_0\|\mathbf{B}\|_F > 0$, so that $x := d_{i^*}$ obeys $x^2 - ax - (\text{OPT}^2 + a\text{OPT} + 2a\|\mathbf{B}\|_F) \leq 0$. Since the leading coefficient is positive, x is at most the larger root of the associated quadratic, namely

$$x \leq \frac{a}{2} + \sqrt{\frac{a^2}{4} + \text{OPT}^2 + a\text{OPT} + 2a\|\mathbf{B}\|_F} = \frac{a}{2} + \sqrt{u^2 + v},$$

where we have introduced the notation $u := \text{OPT} + \frac{a}{2} > 0$ and $v := 2a\|\mathbf{B}\|_F > 0$. Now $(u + \frac{v}{2u})^2 = u^2 + v + \frac{v^2}{4u^2} \geq u^2 + v$, and hence $\sqrt{u^2 + v} \leq u + \frac{v}{2u}$. Using this, we obtain a bound

$$x \leq \frac{a}{2} + u + \frac{v}{2u} = \frac{a}{2} + \left(\text{OPT} + \frac{a}{2}\right) + \frac{a\|\mathbf{B}\|_F}{\text{OPT} + a/2} = \text{OPT} + a + \frac{a\|\mathbf{B}\|_F}{\text{OPT} + a/2}.$$

Discarding the term $a/2 \geq 0$ from the last denominator gives

$$d_{i^*} \leq \text{OPT} + a\left(1 + \frac{\|\mathbf{B}\|_F}{\text{OPT}}\right) = \text{OPT} + 2\varepsilon_0\|\mathbf{B}\|_F\left(1 + \frac{\|\mathbf{B}\|_F}{\text{OPT}}\right). \quad (8)$$

Step 3: the warm start. By assumption $\|\mathbf{B}\|_F \leq \gamma\text{OPT}$ so by our choice of $\varepsilon_0 = \varepsilon/(2\gamma(1+\gamma))$,

$$d_{i^*} \leq \text{OPT} + 2\varepsilon_0\gamma(1+\gamma)\text{OPT} = (1+\varepsilon)\text{OPT}. \quad \square$$

4 The estimation routine

The routine below is the workhorse, invoked with a pivot $\bar{\mathbf{A}}$, an accuracy $\varepsilon_0 \in (0, 1]$, and a failure probability δ . Its structure is that of a Hutch++-style estimator: deflate a low-rank part, evaluate its contribution exactly, and sketch only the flat remainder. It differs in where the deflation is applied. A direct transcription would deflate each product $\mathbf{B}_i^\top \mathbf{B}$, whose trace is the target $\langle \mathbf{B}, \mathbf{B}_i \rangle$; that would require a separate rangefinder per candidate, and hence $\Omega(N)$ queries. Instead we deflate the single unknown factor $\mathbf{B}$ once, at a cost of $O(k)$ queries shared across all N candidates, and flatten the other factor $\mathbf{B}_i$ offline by truncation — which is free, because $\mathbf{B}_i$ is known. The product $\mathbf{N}_i = (\mathbf{B}_i^H)^\top \mathbf{R}$ is then flat because both of its factors are, which is what the union bound over N candidates needs.

Algorithm EstimateScores($\bar{\mathbf{A}}, \varepsilon_0, \delta$). Set

$$k := \left\lceil C(\varepsilon_0^{-2}\sqrt{\log(N/\delta)} + \log(1/\delta)) \right\rceil, \quad r_0 := \lceil \alpha \varepsilon_0^2 k \rceil, \quad (9)$$

where C is a sufficiently large absolute constant and $\alpha \in (0, 1)$ is a sufficiently small one. Thus $k = O(\varepsilon_0^{-2}\sqrt{\log(N/\delta)} + \log(1/\delta))$ and $r_0 = \Theta(\varepsilon_0^2 k)$. The truncation rank r_0 trades the two error terms against each other: shrinking α improves the low-rank bound ?? and worsens the flat bound, and the analysis fixes α first, then takes C large.

1. **(Sketch.)** Draw $\mathbf{\Omega}, \mathbf{Y} \in \mathbb{R}^{n \times k}$ with i.i.d. $\mathcal{N}(0, 1)$ entries, independent of each other, and issue the $2k$ right queries $\mathbf{A}\mathbf{\Omega}$ and $\mathbf{A}\mathbf{Y}$. Form $\mathbf{B}\mathbf{\Omega} = \mathbf{A}\mathbf{\Omega} - \bar{\mathbf{A}}\mathbf{\Omega}$ and $\mathbf{Q} = \text{orth}(\mathbf{B}\mathbf{\Omega}) \in \mathbb{R}^{n \times k'}$, $k' \leq k$.
2. **(Deflate.)** Issue the k' left queries $\mathbf{A}^\top \mathbf{q}_j$, $j \in [k']$, and form $\mathbf{W} := \mathbf{Q}^\top \mathbf{B} = (\mathbf{A}^\top \mathbf{Q})^\top - \mathbf{Q}^\top \bar{\mathbf{A}}$ and $\hat{\mathbf{B}} := \mathbf{Q}\mathbf{W} = \mathbf{Q}\mathbf{Q}^\top \mathbf{B}$. Write $\mathbf{R} := \mathbf{B} - \hat{\mathbf{B}} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{B}$ (unknown to the algorithm).
3. **(Split the candidates.)** For each i , compute the SVD of $\mathbf{B}_i$ and split

$$\mathbf{B}_i = \mathbf{B}_i^L + \mathbf{B}_i^H, \quad \mathbf{B}_i^L := [\mathbf{B}_i]_{r_0},$$

so that $\text{rank}(\mathbf{B}_i^L) \leq r_0$ and $\|\mathbf{B}_i^H\|_2 = \sigma_{r_0+1}(\mathbf{B}_i) \leq \|\mathbf{B}_i\|_F / \sqrt{r_0 + 1}$.

4. **(Quadratic sketch.)** Using the queries $\mathbf{A}\mathbf{Y}$ already made in step 1, form $\mathbf{R}\mathbf{Y} = (\mathbf{A}\mathbf{Y} - \bar{\mathbf{A}}\mathbf{Y}) - \hat{\mathbf{B}}\mathbf{Y}$ and, for each i ,

$$\hat{\psi}_i := \frac{1}{k} \langle \mathbf{B}_i^H \mathbf{Y}, \mathbf{R}\mathbf{Y} \rangle = \frac{1}{k} \sum_{b=1}^k \mathbf{y}_b^\top (\mathbf{B}_i^H)^\top \mathbf{R} \mathbf{y}_b.$$

5. **(Output.)** Return $\hat{\varphi}_i := \langle \mathbf{B}_i, \hat{\mathbf{B}} \rangle + \hat{\psi}_i$ for all $i \in [N]$.

The query cost is $2k$ right queries and $k' \leq k$ left queries, i.e. $O(k)$ in total, issued in two rounds: the right queries of step 1 use vectors drawn independently of $\mathbf{A}$, and only the left queries of step 2 depend on answers already received. Everything else — $\bar{\mathbf{A}}\mathbf{\Omega}$, $\mathbf{Q}^\top \bar{\mathbf{A}}$, $\mathbf{A}\mathbf{Y}$, the SVDs of the $\mathbf{B}_i$, and $\mathbf{B}_i^H \mathbf{Y}$ — is computed offline. In particular $\mathbf{R}\mathbf{Y}$ is computable even though $\mathbf{R}$ is not.

Lemma 4.1 (Guarantee for ESTIMATESCORES). *For a sufficiently large absolute constant C in ??, the following holds. For any pivot $\bar{\mathbf{A}}$, any $\varepsilon_0 \in (0, 1]$, and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ the output of ESTIMATESCORES satisfies*

$$|\hat{\varphi}_i - \langle \mathbf{B}, \mathbf{B}_i \rangle| \leq \varepsilon_0 \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F \quad \text{simultaneously for every } i \in [N].$$

Proof. Degenerate cases. If $\|\mathbf{B}\|_F = 0$ then $\mathbf{B}\mathbf{\Omega} = \mathbf{0}$, so $k' = 0$, $\hat{\mathbf{B}} = \mathbf{0}$, $\mathbf{R} = \mathbf{0}$, $\hat{\psi}_i = 0$, and $\hat{\varphi}_i = 0 = \langle \mathbf{B}, \mathbf{B}_i \rangle$ for all i ; the claim holds with probability 1. Similarly, for any index i with $\|\mathbf{B}_i\|_F = 0$ we have $\mathbf{B}_i = \mathbf{0}$, hence $\mathbf{B}_i^L = \mathbf{B}_i^H = \mathbf{0}$, hence $\hat{\varphi}_i = 0 = \langle \mathbf{B}, \mathbf{B}_i \rangle$; such indices may be ignored below. So assume $\|\mathbf{B}\|_F > 0$ and restrict attention to i with $\|\mathbf{B}_i\|_F > 0$.

Deterministic facts. Because $\mathbf{Q}$ has orthonormal columns, $\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top$ is an orthogonal projector, so $\mathbf{R} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{B}$ satisfies

$$\|\mathbf{R}\|_F \leq \|\mathbf{B}\|_F \tag{10}$$

for every realization of $\mathbf{\Omega}$. Also, $\|\mathbf{B}_i^L\|_F \leq \|\mathbf{B}_i\|_F$ and $\|\mathbf{B}_i^H\|_F \leq \|\mathbf{B}_i\|_F$.

Error decomposition. Since $\mathbf{B} = \hat{\mathbf{B}} + \mathbf{R}$ and $\mathbf{B}_i = \mathbf{B}_i^L + \mathbf{B}_i^H$,

$$\langle \mathbf{B}, \mathbf{B}_i \rangle - \langle \mathbf{B}_i, \hat{\mathbf{B}} \rangle = \langle \mathbf{B}_i, \mathbf{R} \rangle = \langle \mathbf{B}_i^L, \mathbf{R} \rangle + \langle \mathbf{B}_i^H, \mathbf{R} \rangle,$$

and therefore, deterministically,

$$|\hat{\varphi}_i - \langle \mathbf{B}, \mathbf{B}_i \rangle| = |(\hat{\psi}_i - \langle \mathbf{B}_i^H, \mathbf{R} \rangle) - \langle \mathbf{B}_i^L, \mathbf{R} \rangle| \leq |\hat{\psi}_i - \langle \mathbf{B}_i^H, \mathbf{R} \rangle| + |\langle \mathbf{B}_i^L, \mathbf{R} \rangle|. \tag{11}$$

The two brackets are bounded separately.

Deflation event. Let E_1 be the event that ?? holds, i.e. $\|\mathbf{R}\|_2 = O(\|\mathbf{B}\|_F/\sqrt{k})$. Since $k = \Omega(\log(1/\delta))$ by ??, ?? gives $\Pr[E_1^c] \leq e^{-\Omega(k)} \leq \delta/4$ for C large enough.

Low-rank part: deterministic given E_1 . Since $\text{rank}(\mathbf{B}_i^L) \leq r_0$,

$$|\langle \mathbf{B}_i^L, \mathbf{R} \rangle| \leq \|\mathbf{B}_i^L\|_* \|\mathbf{R}\|_2 = O\left(\sqrt{\frac{r_0}{k}}\right) \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F \quad \text{on } E_1.$$

By ??, $\sqrt{r_0/k} = O(\sqrt{\alpha} \varepsilon_0)$, so choosing α a small enough absolute constant gives

$$|\langle \mathbf{B}_i^L, \mathbf{R} \rangle| \leq \frac{\varepsilon_0}{4} \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F \quad \text{on } E_1, \text{ for every } i. \quad (12)$$

Flat part: Hanson–Wright. Condition on $\mathbf{\Omega}$; then $\mathbf{Q}$, $\mathbf{R}$, and every $\mathbf{B}_i^H$ are fixed, and $\mathbf{Y}$ is an independent Gaussian matrix. Each $\hat{\psi}_i$ is an averaged quadratic form with matrix $\mathbf{N}_i := (\mathbf{B}_i^H)^\top \mathbf{R}$ and mean $\text{tr}(\mathbf{N}_i) = \langle \mathbf{B}_i^H, \mathbf{R} \rangle$. Because $r_0 + 1 \geq \alpha \varepsilon_0^2 k$ we have $\|\mathbf{B}_i^H\|_2 \leq \|\mathbf{B}_i\|_F/\sqrt{r_0 + 1} = O(\|\mathbf{B}_i\|_F/(\varepsilon_0 \sqrt{k}))$, the implied constant depending only on the now-fixed α ; so on E_1 ,

$$\|\mathbf{N}_i\|_F \leq \|\mathbf{B}_i^H\|_2 \|\mathbf{R}\|_F = O\left(\frac{\|\mathbf{B}_i\|_F \|\mathbf{B}\|_F}{\varepsilon_0 \sqrt{k}}\right), \quad \|\mathbf{N}_i\|_2 \leq \|\mathbf{B}_i^H\|_2 \|\mathbf{R}\|_2 = O\left(\frac{\|\mathbf{B}_i\|_F \|\mathbf{B}\|_F}{\varepsilon_0 k}\right),$$

using $\|\mathbf{R}\|_F \leq \|\mathbf{B}\|_F$ from ?? for the first bound. Apply ?? with $t_i := \frac{\varepsilon_0}{2} \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F > 0$; the two exponents satisfy

$$\frac{kt_i^2}{\|\mathbf{N}_i\|_F^2} = \Omega(\varepsilon_0^4 k^2), \quad \frac{kt_i}{\|\mathbf{N}_i\|_2} = \Omega(\varepsilon_0^2 k^2) = \Omega(\varepsilon_0^4 k^2),$$

the last step because $\varepsilon_0 \leq 1$. Since $\varepsilon_0^2 k \geq C\sqrt{\log(N/\delta)}$ by ??, both exponents are $\Omega(C^2 \log(N/\delta))$. With α already fixed, taking C large enough makes the right side of ?? at most $\delta/(4N)$. Thus, for every realization of $\mathbf{\Omega}$ in E_1 , the conditional probability of $|\hat{\psi}_i - \langle \mathbf{B}_i^H, \mathbf{R} \rangle| > t_i$ is at most $\delta/(4N)$, and a union bound over $i \in [N]$ gives

$$\Pr\left[E_1 \cap \bigcup_{i \in [N]} \{|\hat{\psi}_i - \langle \mathbf{B}_i^H, \mathbf{R} \rangle| > t_i\}\right] \leq \frac{\delta}{4}.$$

Conclusion. With probability at least $1 - \Pr[E_1^c] - \delta/4 \geq 1 - \delta/2 \geq 1 - \delta$, both ?? and $|\hat{\psi}_i - \langle \mathbf{B}_i^H, \mathbf{R} \rangle| \leq t_i$ hold for all i , and then ?? gives

$$|\hat{\varphi}_i - \langle \mathbf{B}, \mathbf{B}_i \rangle| \leq \frac{\varepsilon_0}{2} \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F + \frac{\varepsilon_0}{4} \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F \leq \varepsilon_0 \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F. \quad \square$$

Remark 4.2 (Where $\sqrt{\log N}$ comes from). The union bound in the proof is paid against exponents of order $\varepsilon_0^4 k^2$, i.e. against an *effective sample size* of k^2 from only $O(k)$ queries. Both factors of k have distinct origins: one from the averaging over the k columns of $\mathbf{Y}$, and one from the spectral flatness of $\mathbf{N}_i = (\mathbf{B}_i^H)^\top \mathbf{R}$, whose two factors were flattened separately — $\mathbf{B}_i^H$ by the rank- r_0 truncation (legal because $\mathbf{B}_i$ is known), and $\mathbf{R}$ by the rangefinder (legal because left queries make $\mathbf{Q}^\top \mathbf{B}$ exactly computable). Setting $\varepsilon_0^4 k^2 = \Theta(\log N)$ gives $k = \Theta(\varepsilon_0^{-2} \sqrt{\log N})$. Without the two flattening steps the exponent degrades to order $\varepsilon_0^2 k$, forcing $k = \Omega(\varepsilon_0^{-2} \log N)$; see ??.

5 Proof of the main theorem

Proof of ??. Take the pivot $\bar{\mathbf{A}} := \bar{\mathbf{A}}_0$, so that $\mathbf{B} = \mathbf{A} - \bar{\mathbf{A}}_0$ and $\mathbf{B}_i = \mathbf{A}_i - \bar{\mathbf{A}}_0$. Let ε_0 be as in ??, run ESTIMATESCORES($\bar{\mathbf{A}}_0, \varepsilon_0, \delta$), and output $\hat{i} \in \arg \min_i (\|\mathbf{B}_i\|_F^2 - 2\hat{\varphi}_i)$. ?? supplies $\|\mathbf{B}\|_F \leq \gamma \text{OPT}$, so $\bar{\mathbf{A}}_0$ is a legal pivot for ??. By ??, the hypothesis ?? of ?? holds with probability at least $1 - \delta$, and on that event ?? gives $d_i \leq (1 + \varepsilon) \text{OPT}$. The cost is $O(k)$ queries, with

$$k = O(\varepsilon_0^{-2} \sqrt{\log(N/\delta)} + \log(1/\delta)) = O(\gamma^4 \varepsilon^{-2} \sqrt{\log(N/\delta)} + \log(1/\delta)),$$

since $\varepsilon_0^{-2} = 4\gamma^2(1 + \gamma)^2\varepsilon^{-2} = O(\gamma^4\varepsilon^{-2})$, using $1 + \gamma \leq 2\gamma$. $\square$

6 Discussion

6.1 Left queries are necessary, in any bounded-precision model

Fix m and a code $\{\mathbf{v}_1, \dots, \mathbf{v}_N\} \subset \{\pm 1\}^m$ with pairwise Hamming distance at least $m/4$; by Gilbert–Varshamov such a code exists with $m = O(\log N)$. Embed each $\mathbf{v}_i$ in $\mathbb{R}^n$ by zero-padding and take

$$\mathcal{F} := \{\mathbf{e}_1 \mathbf{v}_i^\top\}_{i \in [N]}, \quad \mathbf{A} := \mathbf{e}_1 \mathbf{w}^\top + \sqrt{m} \mathbf{e}_2 \mathbf{e}_3^\top \quad \text{for some codeword } \mathbf{w}.$$

The second term of $\mathbf{A}$ lies outside the span of $\mathcal{F}$ and is there only to keep $\text{OPT} > 0$, as ?? requires. Since $\mathbf{A} - \mathbf{e}_1 \mathbf{v}_i^\top = \mathbf{e}_1 (\mathbf{w} - \mathbf{v}_i)^\top + \sqrt{m} \mathbf{e}_2 \mathbf{e}_3^\top$ has orthogonal summands, and $\|\mathbf{w} - \mathbf{v}_i\|_2^2 = 4 \text{Ham}(\mathbf{w}, \mathbf{v}_i) \geq m$ for $\mathbf{v}_i \neq \mathbf{w}$,

$$d_i^2 = \|\mathbf{w} - \mathbf{v}_i\|_2^2 + m, \quad \text{OPT} = \sqrt{m}, \quad d_i \geq \sqrt{2m} = \sqrt{2} \text{OPT} \quad \text{for } \mathbf{v}_i \neq \mathbf{w}.$$

A $(1 + \varepsilon)$ -approximation with $\varepsilon < \sqrt{2} - 1$ therefore forces exact identification of $\mathbf{w}$. The pivot $\bar{\mathbf{A}}_0 = \mathbf{0}$ is a legal warm start with $\gamma = \sqrt{2}$, since $\|\mathbf{A}\|_F = \sqrt{2m}$, and it carries no information about $\mathbf{w}$.

A right query $\mathbf{y}$ returns $\mathbf{A}\mathbf{y} = (\mathbf{w}^\top \mathbf{y}) \mathbf{e}_1 + \sqrt{m} y_3 \mathbf{e}_2$, whose second component is known in advance. If query entries are restricted to $\{\pm 1, 0\}$ then $\mathbf{w}^\top \mathbf{y} \in \{-m, \dots, m\}$, so each right query has at most $2m + 1$ possible outcomes. An algorithm making T right queries, each chosen as a function of all previous answers, is a decision tree of depth T and branching factor $2m + 1$, and must have at least N leaves; hence

$$T \geq \frac{\log N}{\log(2m + 1)} = \Omega\left(\frac{\log N}{\log \log N}\right).$$

The same bound holds if answers are merely observed to $O(\log m)$ bits of precision. By contrast, a single left query with $\mathbf{x} = \mathbf{e}_1$ returns $\mathbf{A}^\top \mathbf{e}_1 = \mathbf{w}$ in full.

Thus one-sided access costs $\tilde{\Omega}(\log N)$ on instances that two-sided access solves in one query, and this separation persists at the level of the general problem: ?? gives $O(\sqrt{\log N})$ two-sided queries for every instance.

Two caveats. First, with unrestricted real entries and exact arithmetic the lower bound is void: the single right query $\mathbf{y} = (1, 2^{-1}, 2^{-2}, \dots)$ encodes all of $\mathbf{w}$ in one real number. Lower bounds in this model are only meaningful under sign-restricted queries, finite precision, or additive noise; our upper bound, by contrast, survives all of these up to the modifications of ??. Second, the instance shows that any algorithm achieving $o(\log N)$ queries must exploit two-sidedness on *spiky* (low-stable-rank) differences $\mathbf{A} - \mathbf{A}_i$; this is exactly what drives the design of ??.

6.2 Why deflation: the variance obstruction

The natural one-shot estimator for $\langle \mathbf{B}, \mathbf{B}_i \rangle$ is a bilinear sketch, $\frac{1}{k^2} \langle \mathbf{X}^\top \mathbf{B}_i \mathbf{Y}, \mathbf{X}^\top \mathbf{B} \mathbf{Y} \rangle$, or its one-sided cousin $\frac{1}{k} \langle \mathbf{B}_i \mathbf{Y}, \mathbf{B} \mathbf{Y} \rangle$. For the one-sided version, ?? applies with $\mathbf{N}_i = \mathbf{B}_i^\top \mathbf{B}$ and the sub-Gaussian exponent is $kt^2 / \|\mathbf{B}_i^\top \mathbf{B}\|_F^2$. When $\mathbf{B}_i$ and $\mathbf{B}$ are both spiky and share a left factor — say $\mathbf{B}_i = \mathbf{u} \mathbf{a}_i^\top$ and $\mathbf{B} = \mathbf{u} \mathbf{b}^\top$ with $\|\mathbf{u}\|_2 = 1$, which is the situation in ?? with $\mathbf{u} = \mathbf{e}_1$ — one has $\|\mathbf{B}_i^\top \mathbf{B}\|_F = \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F$. At the target accuracy $t = \varepsilon_0 \|\mathbf{B}_i\|_F \|\mathbf{B}\|_F$ the exponent then degrades to $\varepsilon_0^2 k$, and the union bound over N hypotheses forces $k = \Omega(\varepsilon_0^{-2} \log N)$. Equivalently: the effective sample size of a bilinear sketch of a rank-one matrix is k , not k^2 , because $\|\mathbf{X}^\top \mathbf{u} \mathbf{a}^\top \mathbf{Y}\|_F^2 = \|\mathbf{X}^\top \mathbf{u}\|_2^2 \|\mathbf{Y}^\top \mathbf{a}\|_2^2$ is a product of two k -degree-of-freedom factors.

The routine of ?? removes both spiky factors before sketching. The candidate factor is flattened offline, by truncating the known matrix $\mathbf{B}_i$ at rank r_0 ; the truncation error is controlled *deterministically* against $\|\mathbf{R}\|_2$ through the nuclear norm, in ??. The unknown factor is flattened by the rangefinder. It is worth being precise about where left queries enter here. The basis $\mathbf{Q} = \text{orth}(\mathbf{B}\mathbf{\Omega})$ is computable from right queries alone, and ?? is a statement about $\mathbf{Q}$ only; on the rank-one instances of ?? one even has $(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{B} = \mathbf{0}$ exactly. What one-sided access cannot deliver is the deflated matrix $\widehat{\mathbf{B}} = \mathbf{Q}\mathbf{Q}^\top \mathbf{B}$ itself: forming $\mathbf{Q}^\top \mathbf{B}$ is precisely a left-query operation. Without $\widehat{\mathbf{B}}$ the algorithm can neither evaluate the leading term $\langle \mathbf{B}_i, \widehat{\mathbf{B}} \rangle$ of $\widehat{\varphi}_i$ nor form the residual sketch $\mathbf{R}\mathbf{Y}$, and the estimator collapses back to the un-deflated one.

After both flattening steps, both Hanson–Wright branches carry exponents of order $\varepsilon_0^4 k^2$ and $\varepsilon_0^2 k^2$ (??), and the union bound over N hypotheses is absorbed at $k = \Theta(\varepsilon_0^{-2} \sqrt{\log N})$.

6.3 Sign-restricted queries

Suppose query entries must lie in $\{\pm 1, 0\}$, and suppose an a priori bound $\|\mathbf{A}\|_2 \leq \Lambda$ is known. The algorithm survives with two modifications, at the cost of adjusting the absolute constants in ??.

First, the Gaussian sketches $\mathbf{\Omega}$ and $\mathbf{Y}$ are replaced by Rademacher sketches. ?? holds verbatim for Rademacher vectors [rudelson2013hanson]. ?? is a Gaussian statement, and we are not aware of a reference establishing ?? verbatim for Rademacher $\mathbf{\Omega}$; we take the sub-Gaussian analogue as a hypothesis, and flag it as a technical gap.

Second, the left queries with the real-valued columns of $\mathbf{Q}$ are simulated by binary expansion. Writing $\mathbf{q}_j$ to b bits as a signed combination $\tilde{\mathbf{q}}_j = \sum_{a=1}^b 2^{-a} \mathbf{s}_{j,a}$ with $\mathbf{s}_{j,a} \in \{\pm 1, 0\}^n$ costs b legal left queries per column and yields $\|\mathbf{q}_j - \tilde{\mathbf{q}}_j\|_2 \leq 2^{-b} \sqrt{n}$. The algorithm then forms $\widetilde{\mathbf{W}}$ with rows $(\mathbf{A}^\top \tilde{\mathbf{q}}_j)^\top - \mathbf{q}_j^\top \bar{\mathbf{A}}$ and sets $\widehat{\mathbf{B}} := \mathbf{Q} \widetilde{\mathbf{W}}$, so that

$$\mathbf{E} := \widehat{\mathbf{B}} - \mathbf{Q}\mathbf{Q}^\top \mathbf{B}, \quad \|\mathbf{E}\|_F = \|\widetilde{\mathbf{W}} - \mathbf{W}\|_F \leq 2^{-b} \sqrt{nk} \Lambda.$$

Now $\mathbf{R} := \mathbf{B} - \widehat{\mathbf{B}}$ no longer satisfies $\mathbf{R} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{R}$. The splitting identity $\langle \mathbf{B}_i, \mathbf{R} \rangle = \langle \mathbf{B}_i^L, \mathbf{R} \rangle + \langle \mathbf{B}_i^H, \mathbf{R} \rangle$ is unaffected, since $\mathbf{B}_i^L + \mathbf{B}_i^H = \mathbf{B}_i$ holds by construction; the perturbation instead enters the leading term of $\widehat{\varphi}_i$, which now estimates $\langle \mathbf{B}_i, \mathbf{Q}\mathbf{Q}^\top \mathbf{B} \rangle$ with error $\langle \mathbf{B}_i, \mathbf{E} \rangle$, of magnitude at most $\|\mathbf{B}_i\|_F \|\mathbf{E}\|_F$. Likewise $\|\mathbf{R}\|_F \leq \|\mathbf{B}\|_F + \|\mathbf{E}\|_F$ and, on the deflation event, $\|\mathbf{R}\|_2 = O(\|\mathbf{B}\|_F / \sqrt{k}) + \|\mathbf{E}\|_F$. All three perturbations are absorbed once $\|\mathbf{E}\|_F = O(\varepsilon_0 \|\mathbf{B}\|_F)$ with a small enough implied constant.

Setting b requires a lower bound on $\|\mathbf{B}\|_F$, which is obtained from the right queries already made: $\widehat{\beta}^2 := \frac{1}{k} \|\mathbf{B}\mathbf{\Omega}\|_F^2$ is an averaged quadratic form with matrix $\mathbf{B}^\top \mathbf{B}$, mean $\|\mathbf{B}\|_F^2$, $\|\mathbf{B}^\top \mathbf{B}\|_F \leq \|\mathbf{B}\|_2 \|\mathbf{B}\|_F \leq \|\mathbf{B}\|_F^2$, and $\|\mathbf{B}^\top \mathbf{B}\|_2 \leq \|\mathbf{B}\|_F^2/2$, so ?? with $t = \|\mathbf{B}\|_F^2/2$

gives $\frac{1}{2}\|\mathbf{B}\|_{\mathbf{F}}^2 \leq \widehat{\beta}^2 \leq 2\|\mathbf{B}\|_{\mathbf{F}}^2$ except with probability $e^{-\Omega(k)}$. (If $\widehat{\beta} = 0$ then $\mathbf{B}\boldsymbol{\Omega} = \mathbf{0}$, which for $\|\mathbf{B}\|_{\mathbf{F}} > 0$ has probability zero; the algorithm outputs $\arg \min_i \|\mathbf{B}_i\|_{\mathbf{F}}$.) Taking $b = O(\log(nk\Lambda/(\varepsilon_0\|\mathbf{B}\|_{\mathbf{F}})))$ suffices.

The net effect is a multiplicative $O(\log(n\Lambda/(\varepsilon \text{OPT})))$ overhead on the left queries, which are one third of the total; the right-query count is unchanged.

6.4 Computational cost

The query count is the complexity measure here, but the offline work is nontrivial: the algorithm computes a top- r_0 SVD of $\mathbf{B}_i$ for every $i \in [N]$. Since $r_0 \leq k$, randomized low-rank SVD brings the per-candidate cost to $O(n^2k)$, for $O(Nn^2k)$ in total. No attempt has been made to optimize this.

6.5 Open problems

1. **Is the warm start free?** ?? is the only nonstandard input. A Scheffé-style tournament at constant accuracy would produce a constant-factor pivot, but it requires each pairwise comparison to succeed with confidence $1 - N^{-2}$, and any estimator built from k scalar samples of an *un-deflated* residual concentrates only at rate $e^{-\Theta(k)}$ on spiky instances, forcing $k = \Omega(\log N)$. The deflation that repairs the rate is pivot-relative: its error scale is $\varepsilon_0\|\mathbf{B}_i\|_{\mathbf{F}}\|\mathbf{B}\|_{\mathbf{F}}$, and $\|\mathbf{B}\|_{\mathbf{F}}$ is the *current* pivot's distance, not OPT. Iterated contraction from an arbitrary $\mathbf{A}_1$ therefore costs $O(\sqrt{\log N} \log(2 + \kappa))$ queries with $\kappa = \|\mathbf{A} - \mathbf{A}_1\|_{\mathbf{F}}/\text{OPT}$. Can the $\log \kappa$ be removed in a finite-precision model, giving a scale-free $O(\varepsilon^{-2}\sqrt{\log N})$ algorithm?
2. **Matching lower bound.** We know of no $\Omega(\sqrt{\log N})$ lower bound for two-sided sign-restricted (or finite-precision) queries. The instances of ?? are solved in one two-sided query, and Fano-type arguments leak because each query returns n coordinates.
3. **Dependence on ε .** The error term in ?? is already localized as $\varepsilon_0\|\mathbf{B}_i\|_{\mathbf{F}}\|\mathbf{B}\|_{\mathbf{F}}$, which suggests the ε^{-2} might improve to ε^{-1} by a sharper final round; we do not currently see how.
4. **One-sided complexity.** For right-only queries the truth plausibly is $\Theta(\varepsilon^{-2} \log N)$; the upper bound follows from the un-deflated estimator of ??, while ?? gives $\Omega(\log N / \log \log N)$.